Hepatitis C Virus Frameshifting

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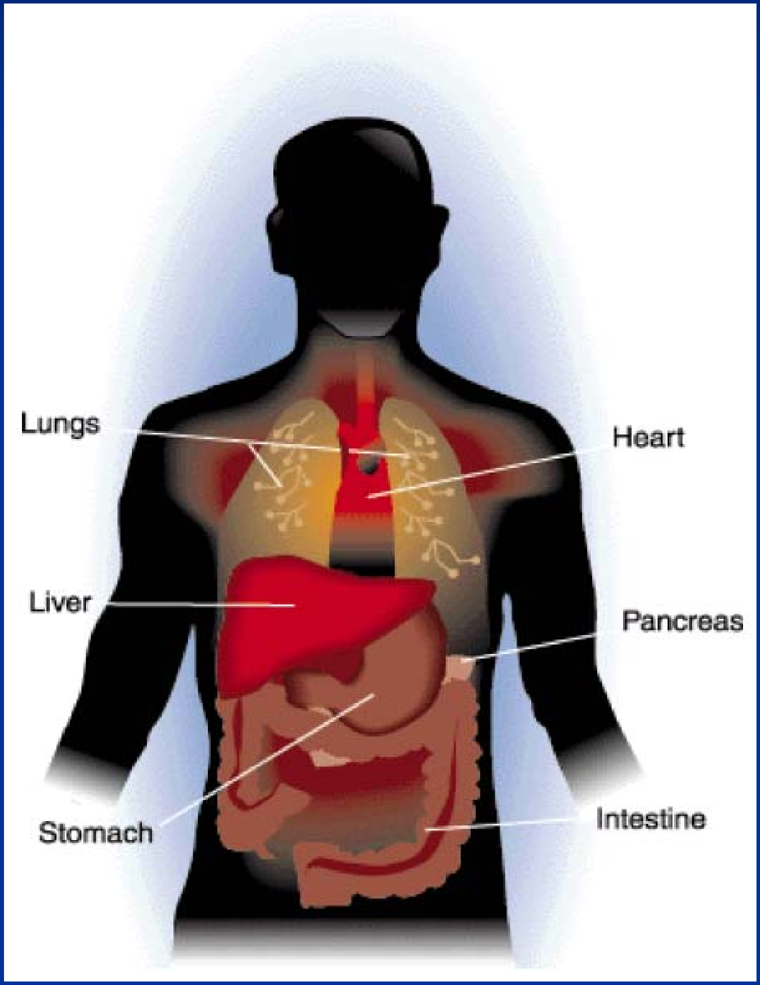
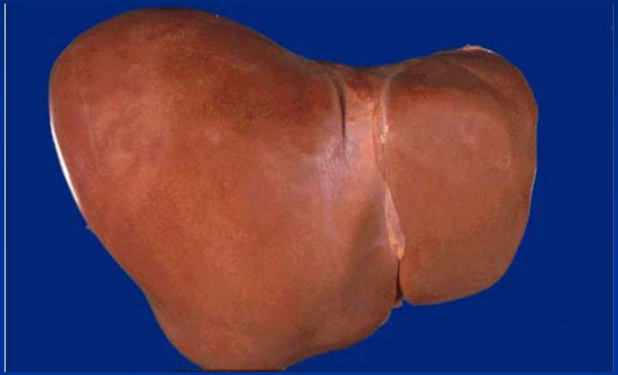
Overview

- Introduction to Hepatitis C Virus
  - Hepatitis
  - HCV Characteristics
- Frameshifting in HCV
  - F-protein
  - Triple code
  - No frameshifting in HCV

Liver





- Regulates blood sugar and amino acids by converting them into usable forms
- Absorbs nutrients from the blood
- Breaks down toxins and drugs
- Manufactures blood proteins
- Produces bile which helps remove waste

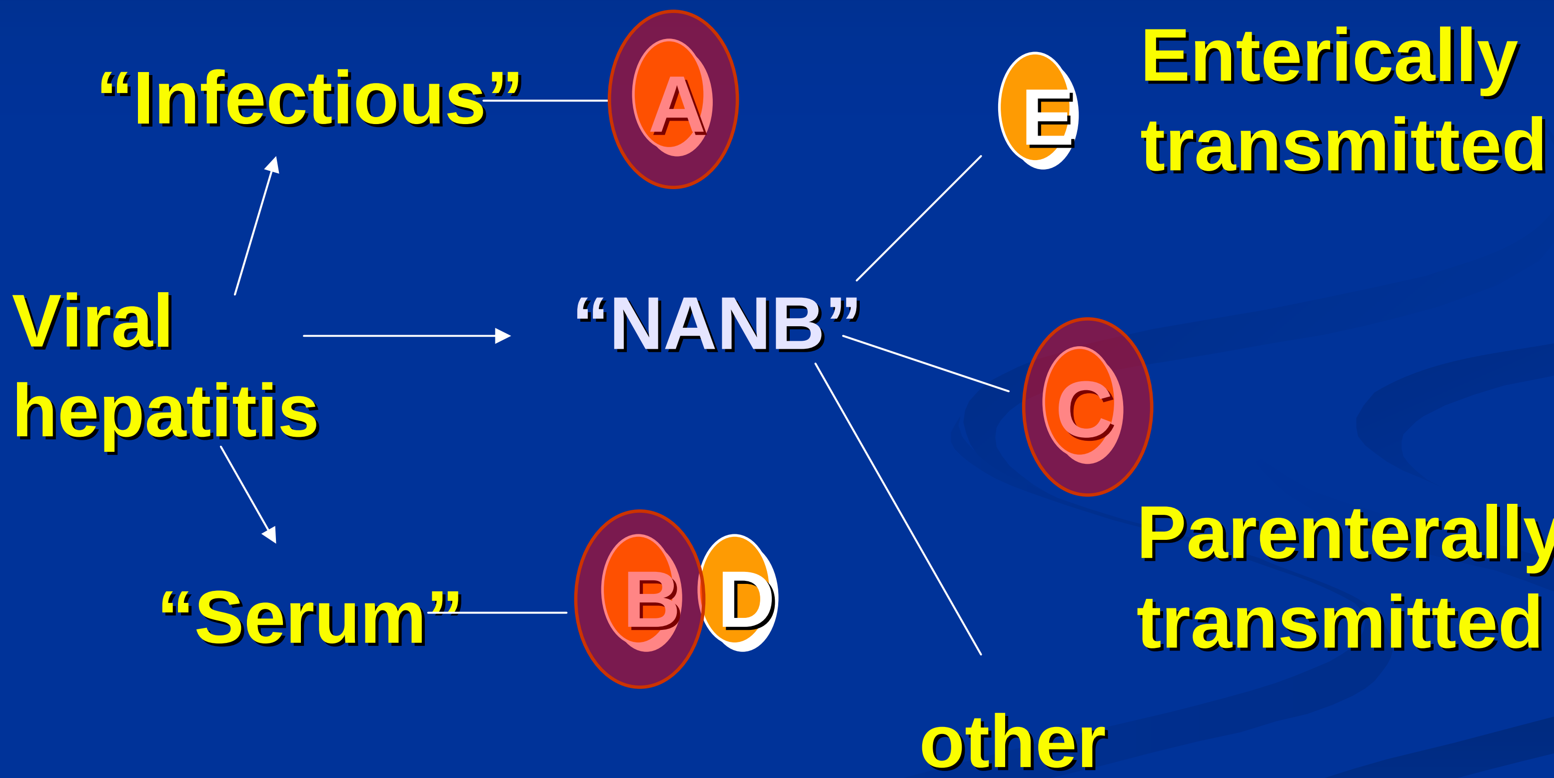
Hepatitis

Hepa – liver itis – inflammation

Hepatitis means inflammation of the liver.

It can be due to a number of causes including: alcohol, medications, chemicals, autoimmune diseases, hereditary or metabolic diseases, and viruses (such as Hepatitis A, B, C, D, E and G).

VIRAL HEPATITIS
HISTORICAL PERSPECTIVE



Estimates of Acute and Chronic Disease Burden for Viral Hepatitis, United States

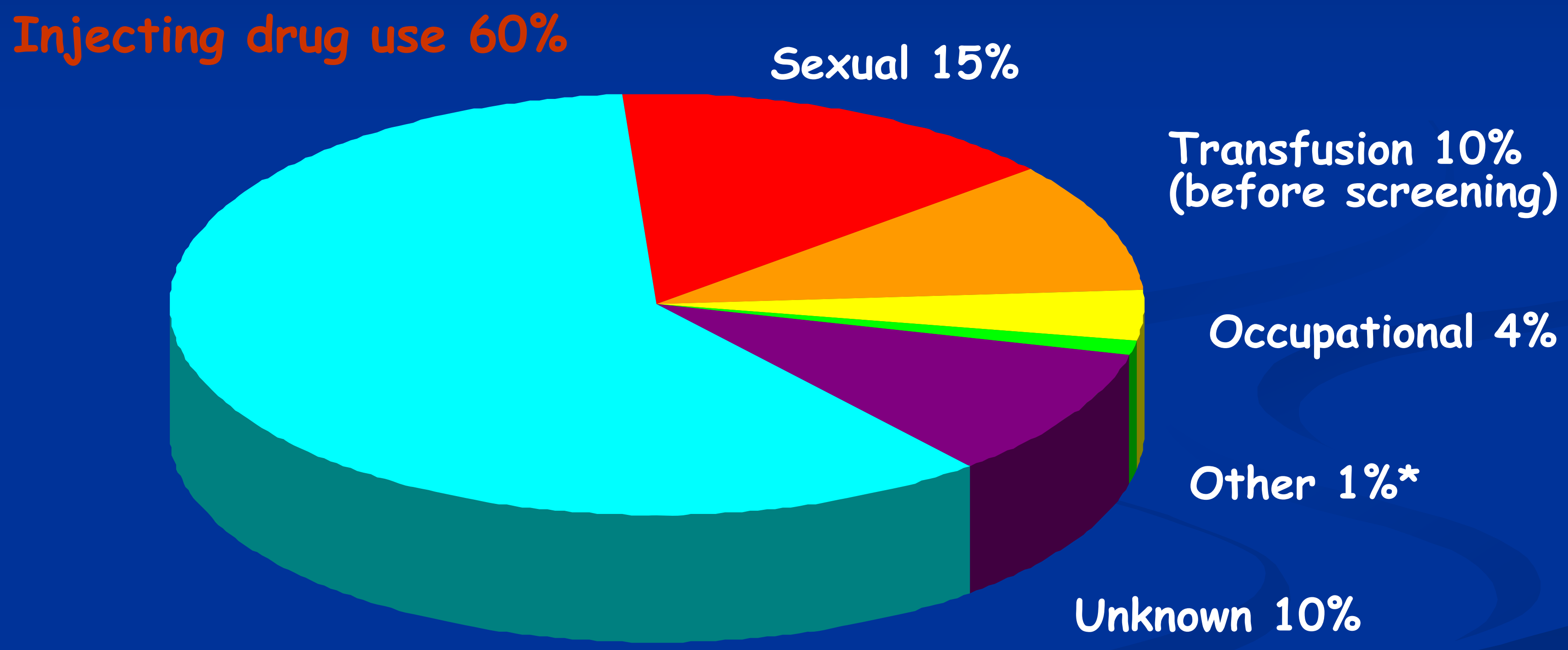
	HAV	HBV	HCV	HDV
Estimated infections (x 1000)/year*	93	78	25	6-13
Fulminant deaths/year	100	150	?	35
Chronic infections	0	1-1.25 million	2.7 million	70,000
Chronic liver disease deaths/year	0	5,000	8-10,000	1,000

* Range based on estimated annual incidence, 2001.

A, B, Cs of Viral Hepatitis

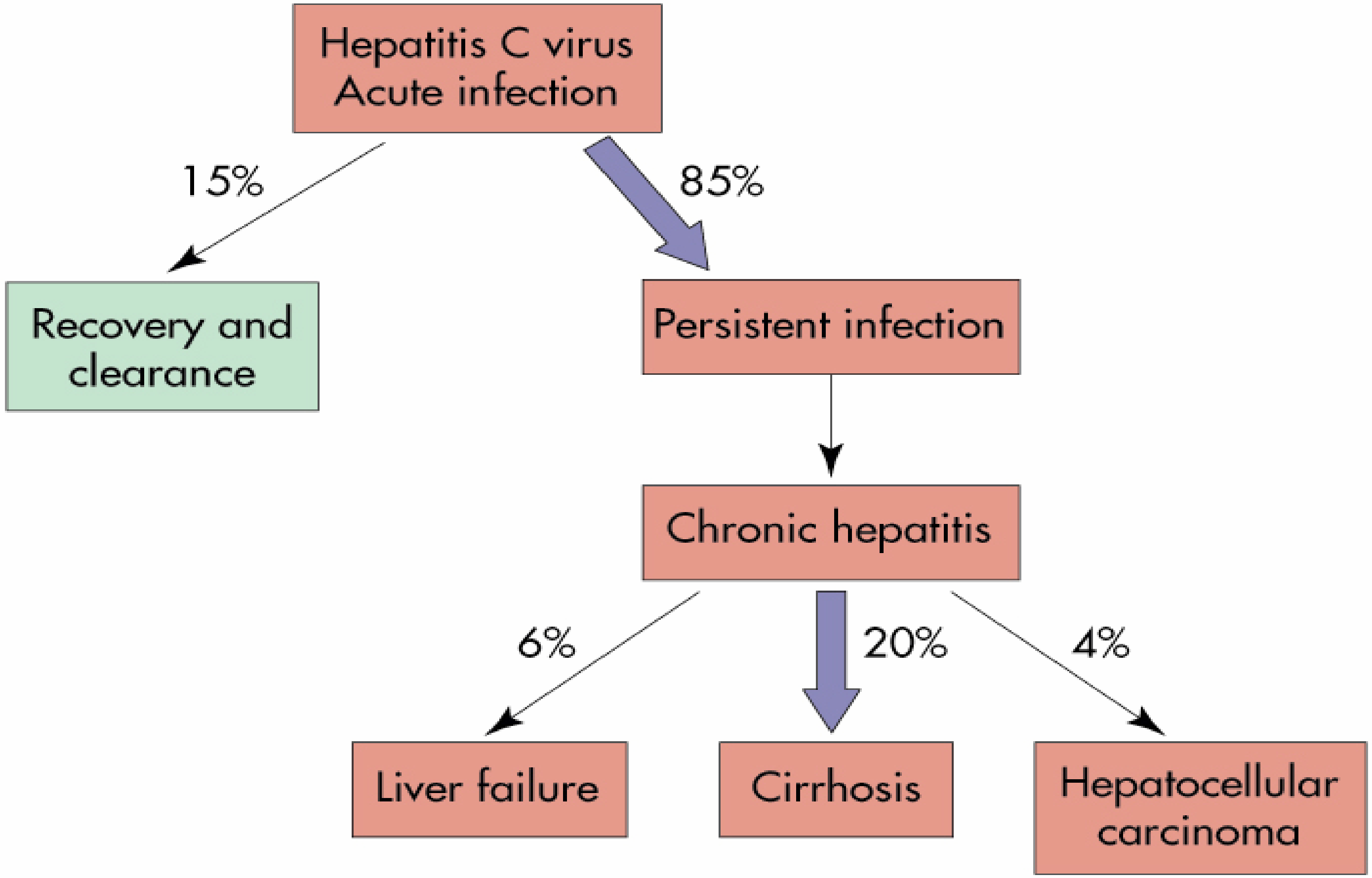
- Hepatitis A
  - fecal-oral spread: hygiene, drug use, men having sex with men, travelers, day care, food
  - vaccine-preventable
- Hepatitis B
  - sexually transmitted – 100x more infectious than HIV
  - Blood borne (sex, injection drug use, mother-child, and health care)
  - vaccine-preventable
- Hepatitis C
  - 200 million people worldwide
  - blood borne (injection drug use primarily)
  - 4-5 times more common than HIV
  - NOT vaccine-preventable!

Sources of Infection for Persons With Hepatitis C



Source: Centers for Disease Control and Prevention

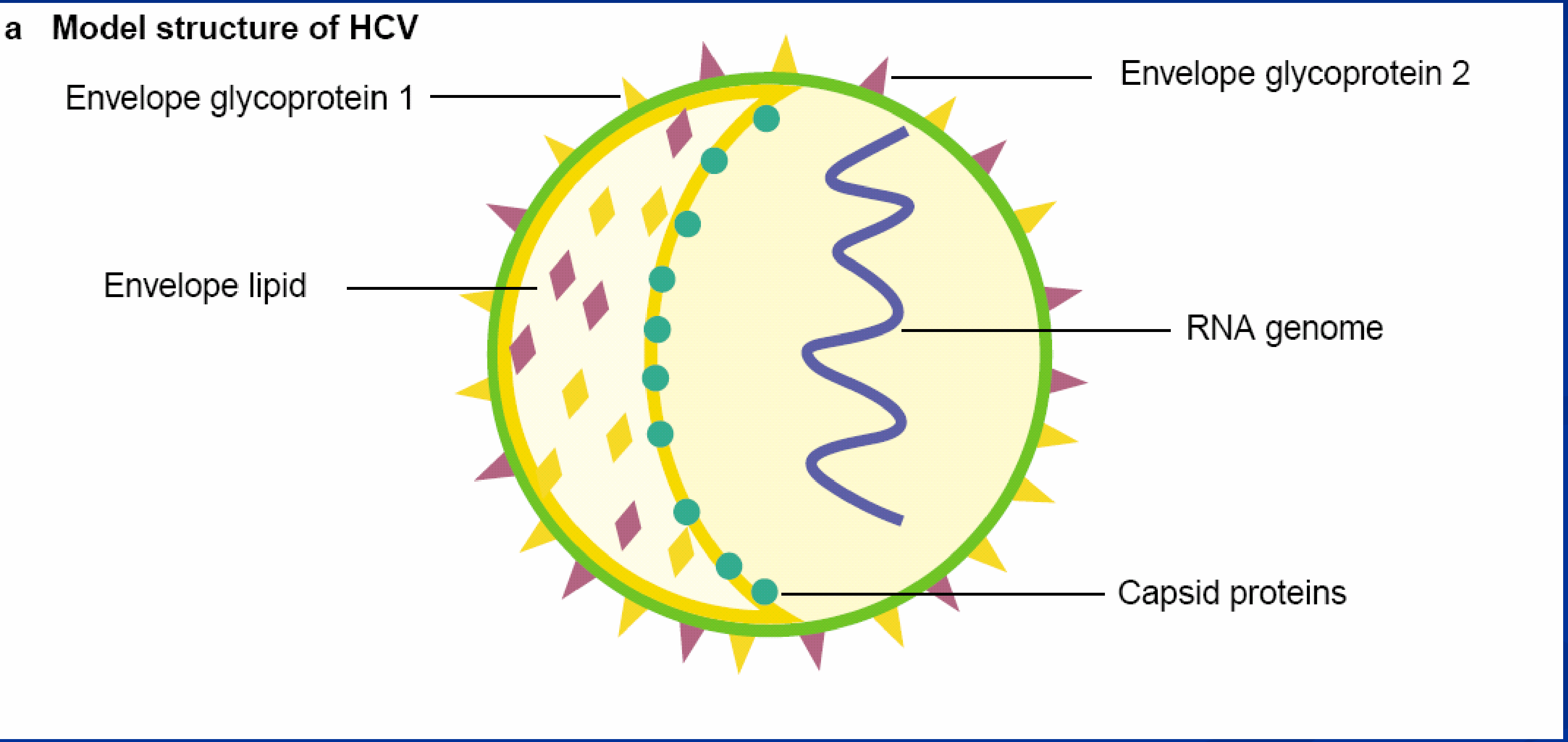
Prognosis of HCV Acute Infection



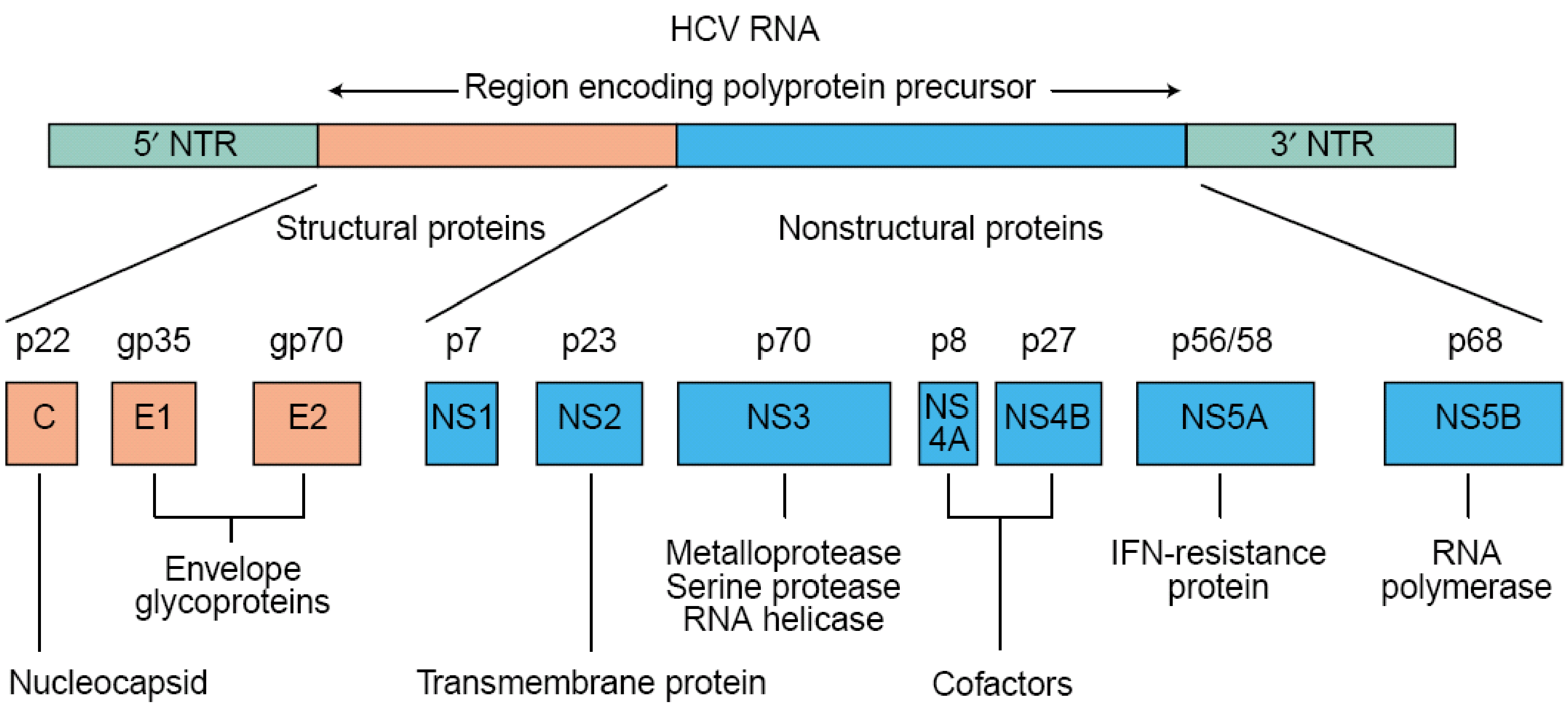
Hepatitis C Viral Characteristic

- Family *Flaviviridae (Hepacivirus)*
- Enveloped
- Positive-sense single-stranded RNA(9.6kb)
- 3,000-amino acid poly-protein (10 proteins)
- No RNA polymerase proofreading ability – **quasispecies**
- $T_{1/2}$: 3 hours (half life)
- Daily production: 12 billion virons

Model Structure of HCV



Protein Encoded by HCV Genomes



Hepatitis C virus (HCV): model structure and genome organisation

History of HCV and F-protein

- First isolated genome HCV-1 in 1989 (Choo et al., 1989)
- 17 kDa Novel protein from the core-protein sequence was discovered both *in vitro* and in mammalian cells (Lo *et al.*, 1994)
- This expression is independent of the downstream E1 envelope protein sequence (Lo *et al.*, 1994, 1995, 1996)
- Role of F protein is still unclear
  - No homologies to other proteins
  - May not be essential for viral RNA replication
  - May involve viral entry, morphology, etc.

History of HCV and F-protein (cont)

- Ribosomal frameshift (Xu *et al.*, 2001, The EMBO Journal)
  +1 shift
- Triple Decoding (Choi *et al.*, 2003, Molecular and Cellular Biology)
  +1 shift // -1 shift
- No frameshift at all. Translation of the F-protein is initiated at a non-AUG codon in a +1 reading frame (Baril and Brakier-Gingras, 2005, Nucleic Acids Research)

The EMBO Journal Vol. 20 No. 14 pp. 3840–3848, 2001

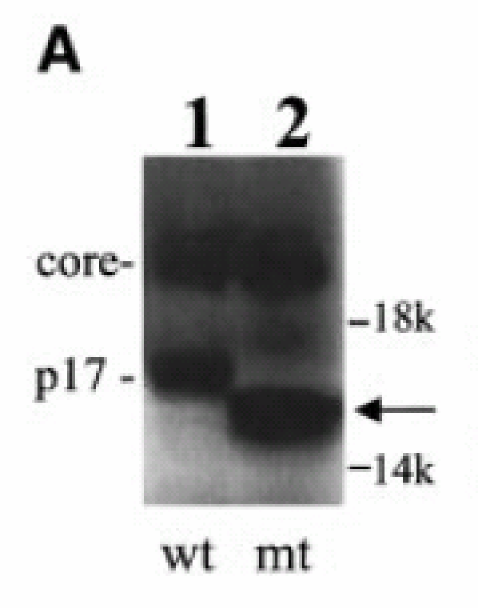
Synthesis of a novel hepatitis C virus protein by ribosomal frameshift

- Inspection of the HCV-1 core protein-coding sequence → overlapping coding sequence in the -2/+1 reading frame.
  - Lack AUG (start) codon near 5′ end
  - Spans from 5 to 485 nucleotide ; coding ~ 160 aas
  - Premature termination mutation at 432 nt results in smaller protein 15kDA
    - F-protein is confirmed to derive from the coding regions of p21 core protein

HCV-1 core protein sequence

```
augagcacgaauccuaaaccucaaaaaaaaaacaaacguaacaccaaccgucgcccacag   60
M  S  T  N  P  K  P  Q  K  K  N  K  R  N  T  N  R  R  P  Q
    A  R  I  L  N  L  K  K  K  T  N  V  T  P  T  V  A  H  R

gacgucaaguucccgggugggcggucagaucguugguggaguuuacuuguugccgcgcagg  120
D  V  K  F  P  G  G  G  Q  I  V  G  G  V  Y  L  L  P  R  R
 T  S  S  S  R  V  A  V  R  S  L  V  E  F  T  C  C  R  A  G

ggcccuagauugggugugcgcgcgacgagaaagacuuccgagcggucgcaaccucgaggu  180
G  P  R  L  G  V  R  A  T  R  K  T  S  E  R  S  Q  P  R  G
 A  L  D  W  V  C  A  R  R  E  R  L  P  S  G  R  N  L  E  V

agacgucagccuauccccaaggcucgucggcccgagggcaggaccugggcucagcccggg  240
R  R  Q  P  I  P  K  A  R  R  P  E  G  R  T  W  A  Q  P  G
 D  V  S  L  S  P  R  L  V  G  P  R  A  G  P  G  L  S  P  G

uacccuuggccccucuauggcaaugagggcugcggguggggcgggauggcuccugucuccc  300
Y  P  W  P  L  Y  G  N  E  G  C  G  W  A  G  W  L  L  S  P
 T  L  G  P  S  M  A  M  R  A  A  G  G  R  D  G  S  C  L  P

cguggcucucggccuagcuggggccccacagacccccggcguaggucgcgcaauuugggu  360
R  G  S  R  P  S  W  G  P  T  D  P  R  R  R  S  R  N  L  G
 V  A  L  G  L  A  G  A  P  Q  T  P  G  V  G  R  A  I  W  V

aaggucaucgauacccuuacgugcggcuucgccgaccucaugggguacauaccgcucguc  420
K  V  I  D  T  L  T  C  G  F  A  D  L  M  G  Y  I  P  L  V
 R  S  S  I  P  L  R  A  A  S  P  T  S  W  G  T  Y  R  S  S

ggcgccccucuuggaggcgcugccagggcccuggcgcauggcguccggguucuggaagac  480
G  A  P  L  G  G  A  A  R  A  L  A  H  G  V  R  V  L  E  D
 A  P  L  L  E  A  L  P  G  P  W  R  M  A  S  G  F  W  K  T

ggcgugaacuaugcaacagggaaccuuccugguugcucuuucucuaucuuccuucuggcc  540
G  V  N  Y  A  T  G  N  L  P  G  C  S  F  S  I  F  L  L  A
 A  *

cugcucucuugcuugacugugcccgcuucggcc                            573
L  L  S  C  L  T  V  P  A  S  A
```

Fig. 1. The HCV core protein sequence and its overlapping coding sequence. The amino acid sequences, shown by one-letter code, were deduced from the sequence of the HCV-1 isolate (Choo *et al.*, 1991). The sequence shown ends at the C-terminus of the core protein-coding sequence. The amino acid sequence encoded by the overlapping coding sequence is shown in bold. Codons 8–14 of the core protein sequence are underlined and nucleotide 432 is boxed.

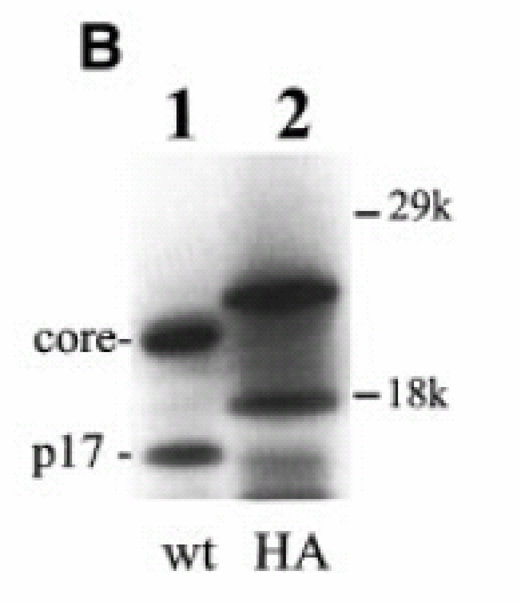


Early termination of +1 frame →
shorter protein

Translation Initiation

- Translation initiation?
  - From a non-AUG codon
  - By frameshift after translation initiation from AUG codon of the core protein seq.
- HA tag, influenza virus hemagglutitin gene, was fused in-frame to the 5' end of the core protein
  - If from non-AUG codon, <u>F-protein should not be affected</u>.
  - If by frameshift, <u>the size of F-protein will increase</u>

Translation Initiation

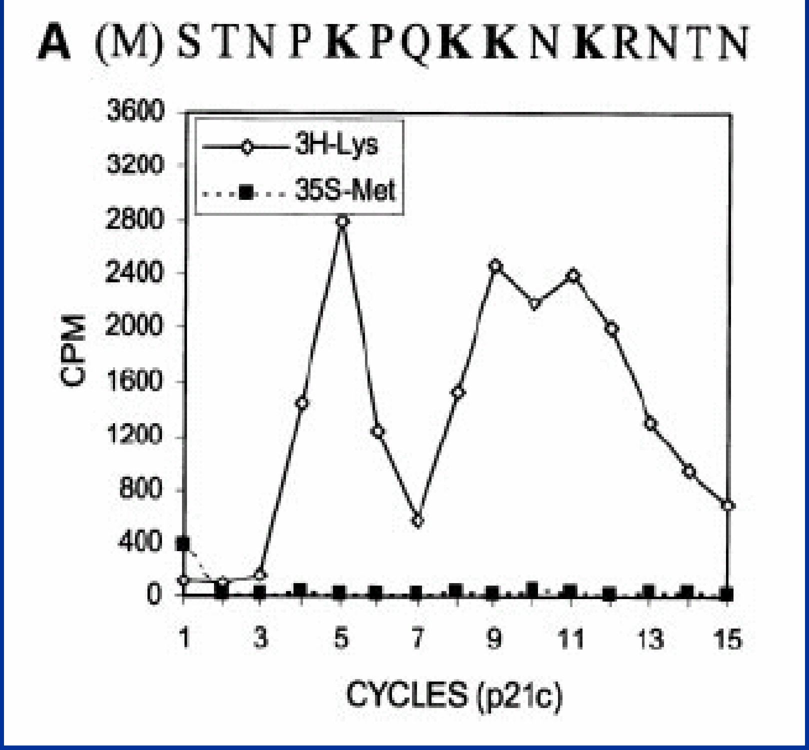


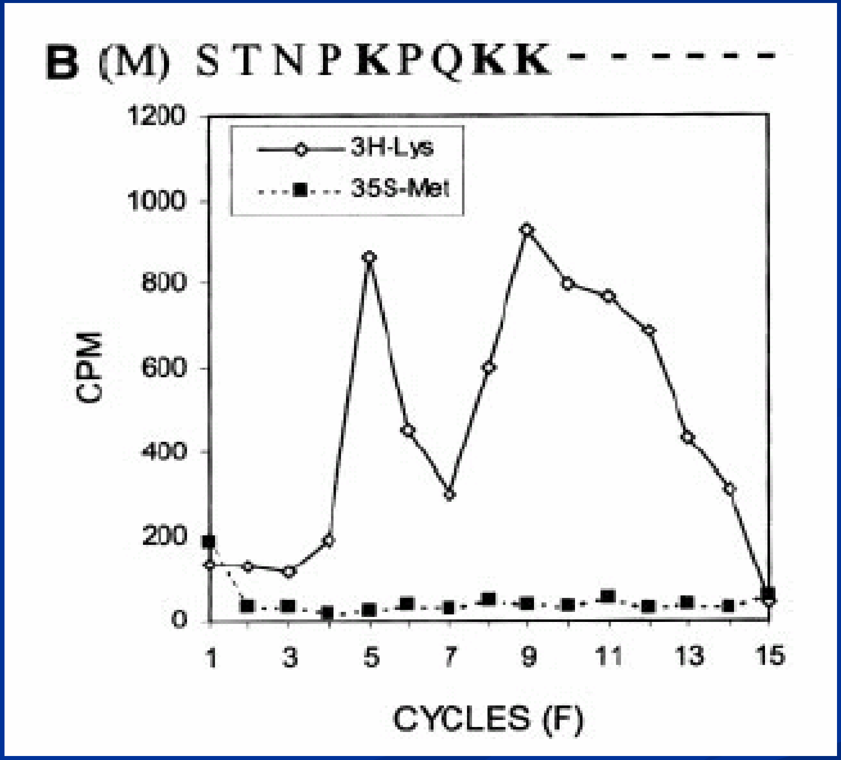
- Sizes of both protein **increased**.
  - ➔17 kDa protein is synthesized from the initiation codon of the core protein seq.
  - ➔Involves frameshift.

Ribosomal Frameshift

- Previously known fact
  - Mutation in codon 9 significantly reduces the efficiency of F protein expression (Lo et al., 1994, 1995)
    - ➔ Frameshift happens in the vicinity of this codon.
- Experimental confirmation
  - Radiosequencing
    - [^{35}S] methionine and [^{3}H]lysine

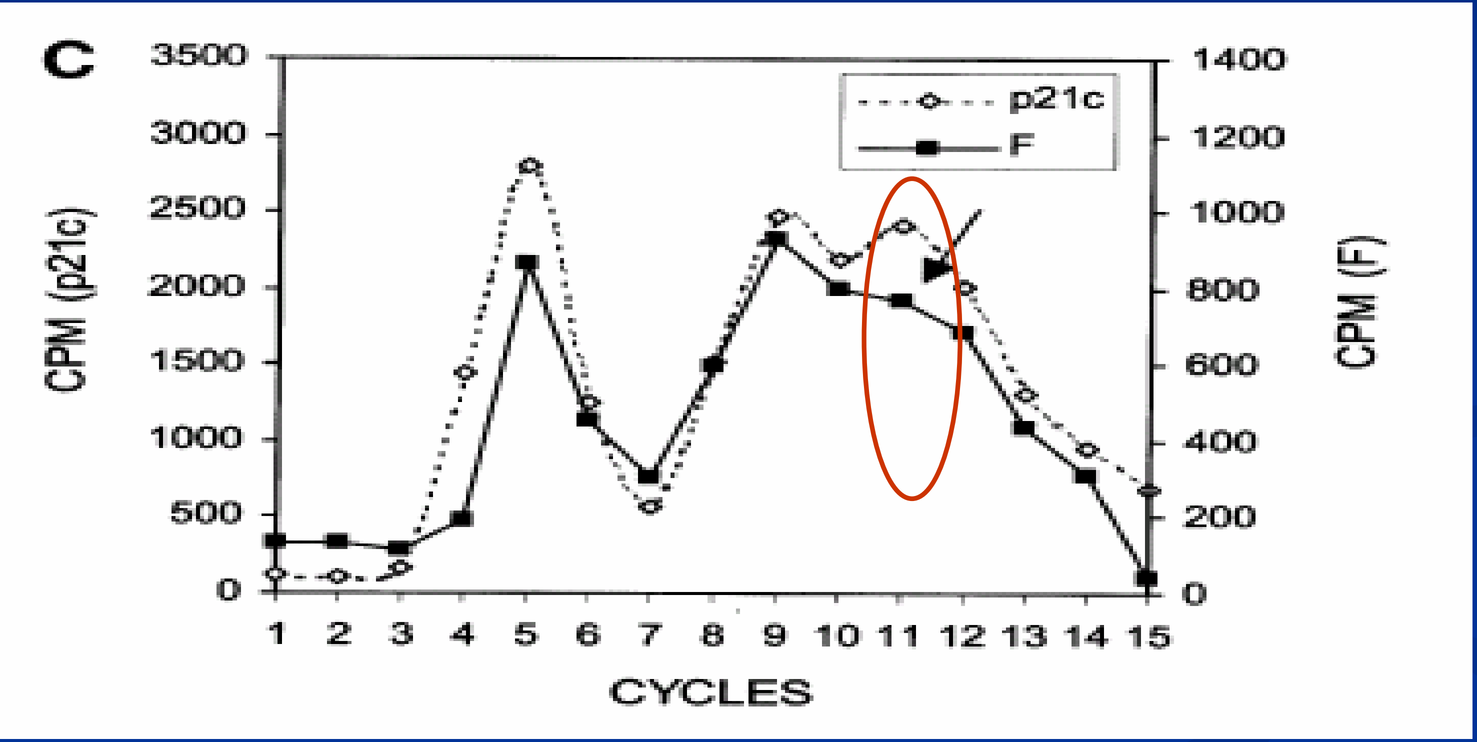
RadioSequencing





Initiator methionine is cleaved away from either proteins

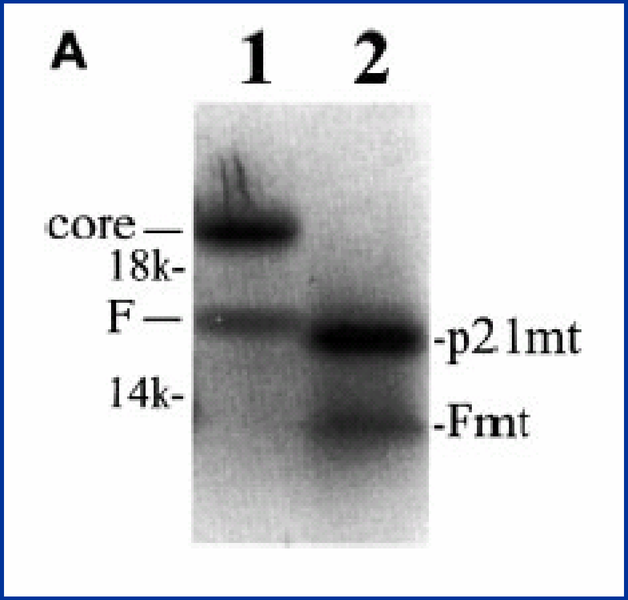
RadioSequencing



F protein diverge at or in the vicinity of codon 11

A-rich sequences

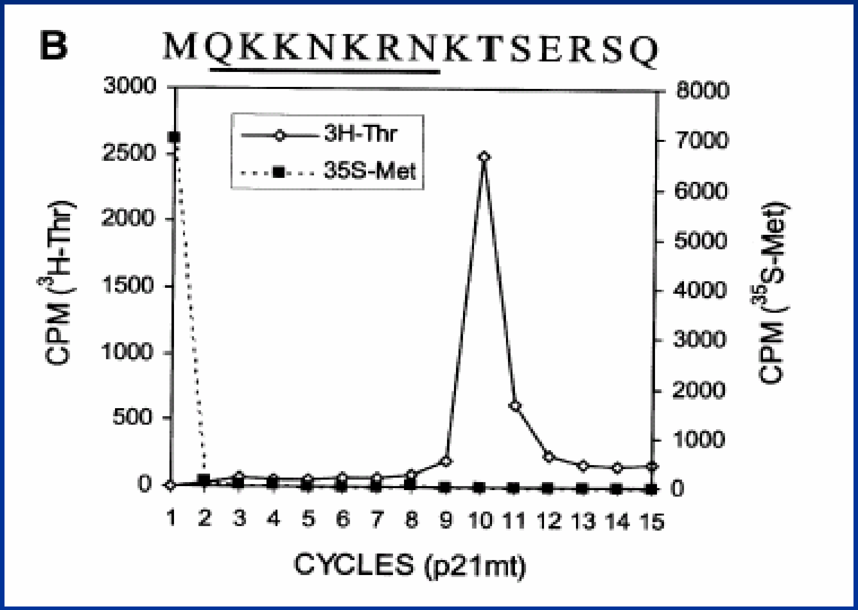
- A-rich region : codons 8-14
- Are these important?
  - Removal of codons 2-7 and 15-50 (flanking the putative ribosomal frameshift site)

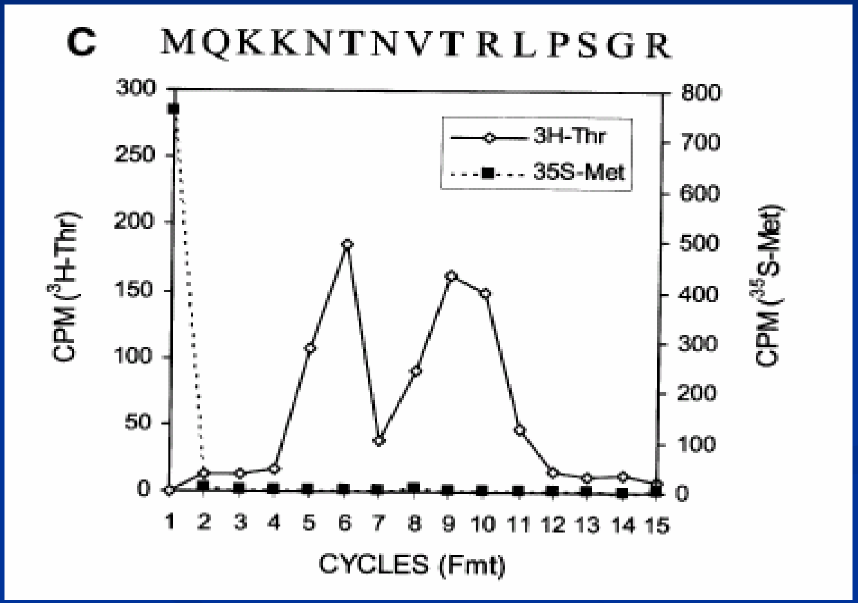


Two smaller protein were detected

→Radiosequencing [^{3}H]threonine and [^{35}S] methionine

A-rich sequences



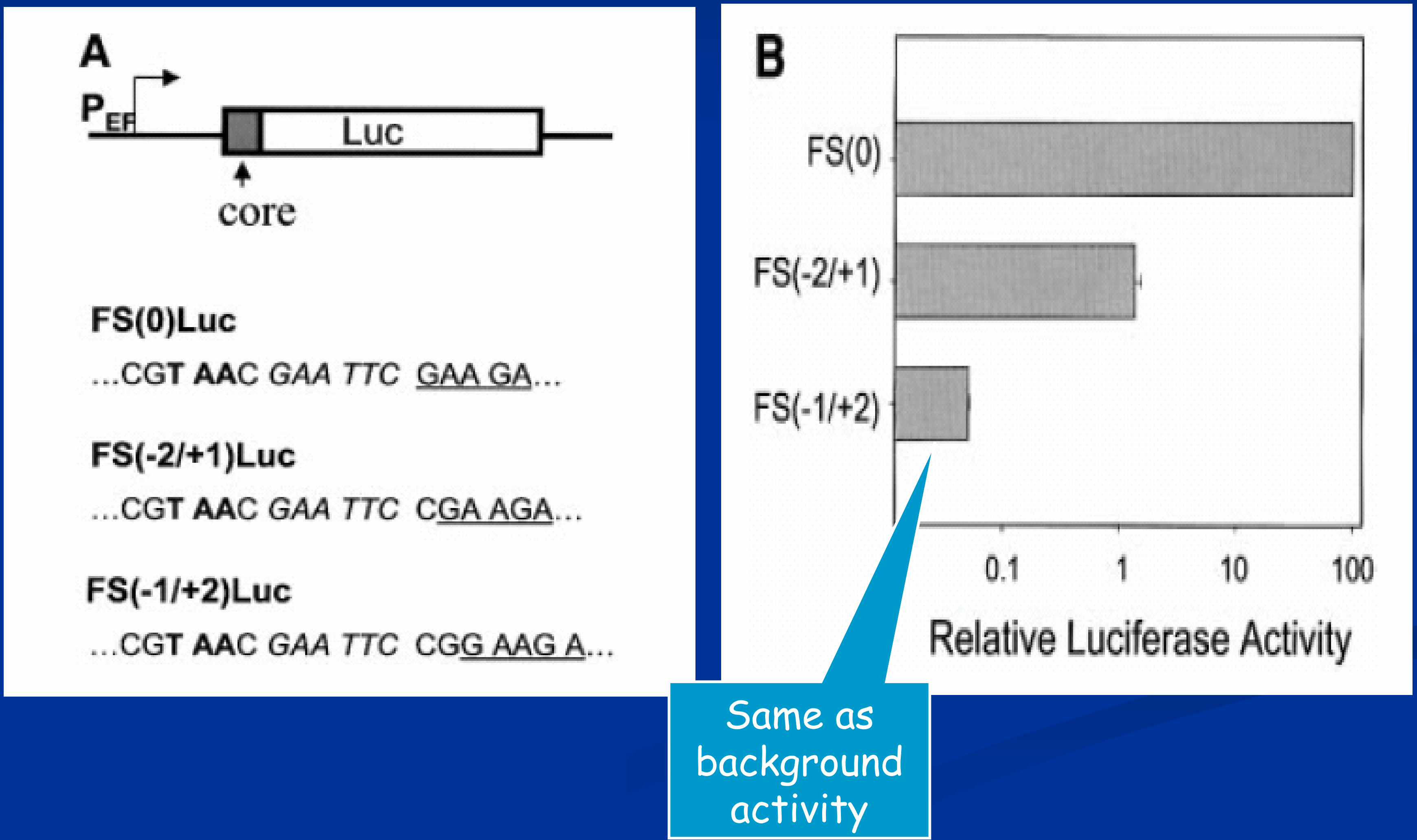


- Prevented the cleavages of the initiation methionine from these two proteins
- The peaks in F protein is broader ➔ possible heterogeneity
  - +1 frameshift : MQKKTNVTR (codon 9,10,11)
    MQKKNNVTR (codon 12)
  - -2 frameshift : MQKKKTNVTR (codon 9,10,11)
    MQKKNTNVTR (codon 12)

Frameshift in Huh7 cells

- Previous results → codons 8-14 were sufficient to mediate ribosomal frameshift *in vitro*
- Then, what about *in vivo*?
- Fusion of the first 14 codons to luciferase-coding sequence.
  - zero frame :
  - -2/+1 frame :
  - -1/+2 frame : negative control. Stop codon in the upstream

Fusion with Luciferase



Conclusion of Xu *et al.*

- F – protein is synthesized by frameshifting from the core protein sequence
- The codon 8-14 plays crucial role in frameshifting
- This frameshifting happens both *in vitro* and *in vivo*

Triple Decoding of Hepatitis C Virus RNA by Programmed Translational Frameshifting

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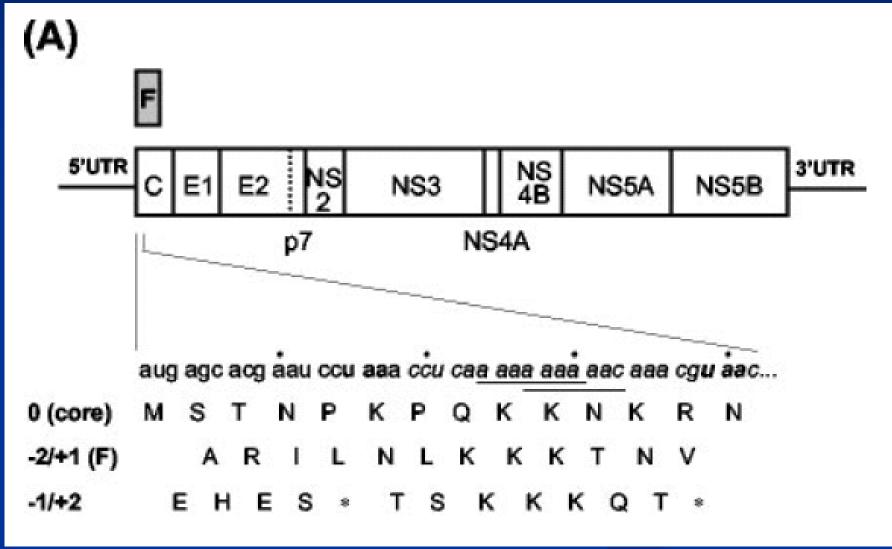
Received 12 June 2002/Returned for modification 21 August 2002/Accepted 27 November 2002

Choi *et al.*

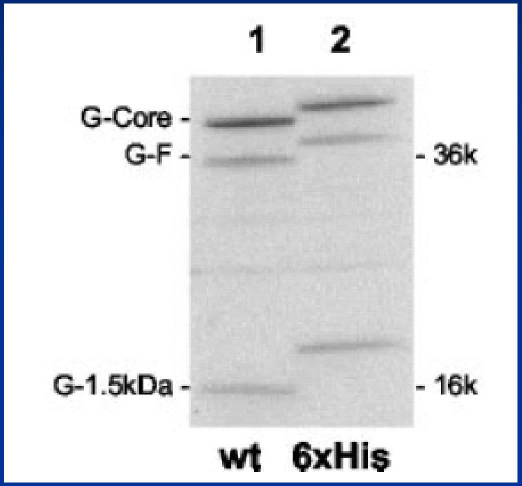
- Frameshift mediates synthesis of three distinct proteins
  - Core protein
  - F protein of the -2/+1 frame
  - 1.5 kDa protein of the -1/+2 frame
- Does not require sequences flanking the frameshift signal
- Two consensus -1 frameshift sequence
- Sequence immediately downstream of the frameshift signal has the potential to form a double stem-loop structure that can significantly enhance frameshifting.

Possible -1/+2 Frameshifting



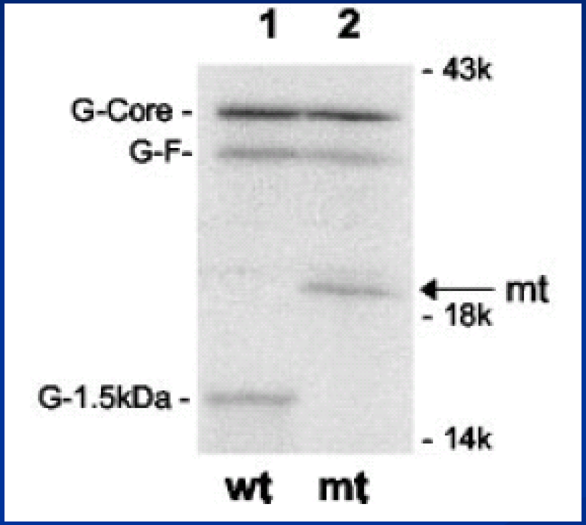
- Contains -1 ribosomal frame shift signals
  - A AAA AAA and A AAA AAC (X XX Y YYZ)
- Possible protein from -1/+2 frameshifting
  - But very small due to early termination at codon 14

Insertion of human a-globin sequence



- Human a-globin sequence (~14 kDa) was fused to the 5' end of the HCV core protein, immediately after the initiation codon.
- Expected size increase
  - P21 core : 21 -> 35
  - F-protein : 17 -> 31
  - 1.5 kDa : 1.5 -> 16.5 → in good agreement

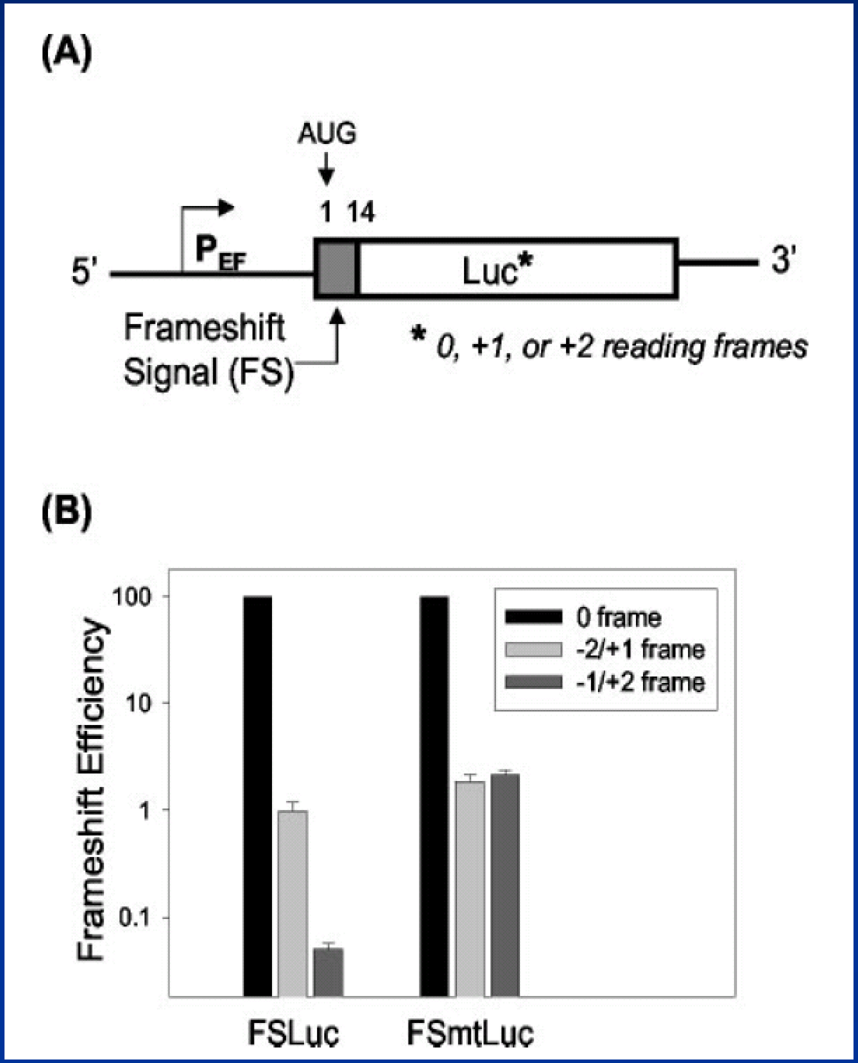
Mutation of codon 14



- Does this protein terminate in the -1/+2 reading frame?
  - Stop codon at 14 and 43
  - Mutate TAA to CAA of codon 14
    - Make translation to continue without stopping at codon 14
    - Should not affect the core and F protein translation

➔ in good agreement

Frameshift in Huh7 cell



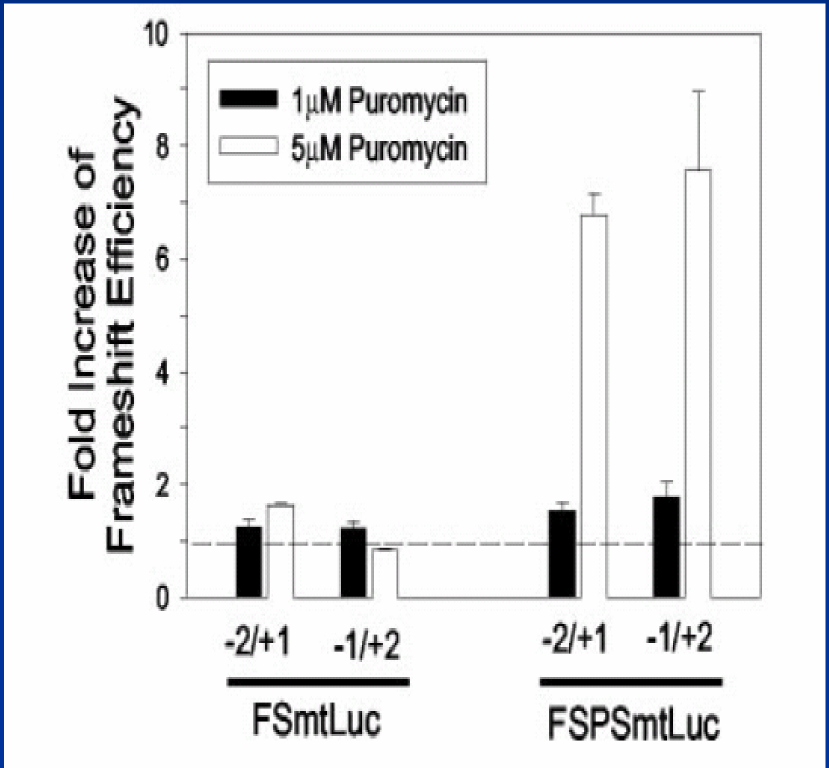
- Fusion of Luciferage gene with codon 1 through 14
- -1/+2 frameshift
  - Previous paper - served as negative control
  - This paper – too short fragment (stop at 14)
- Check if mutation at codon 14 will increase the luciferase activity

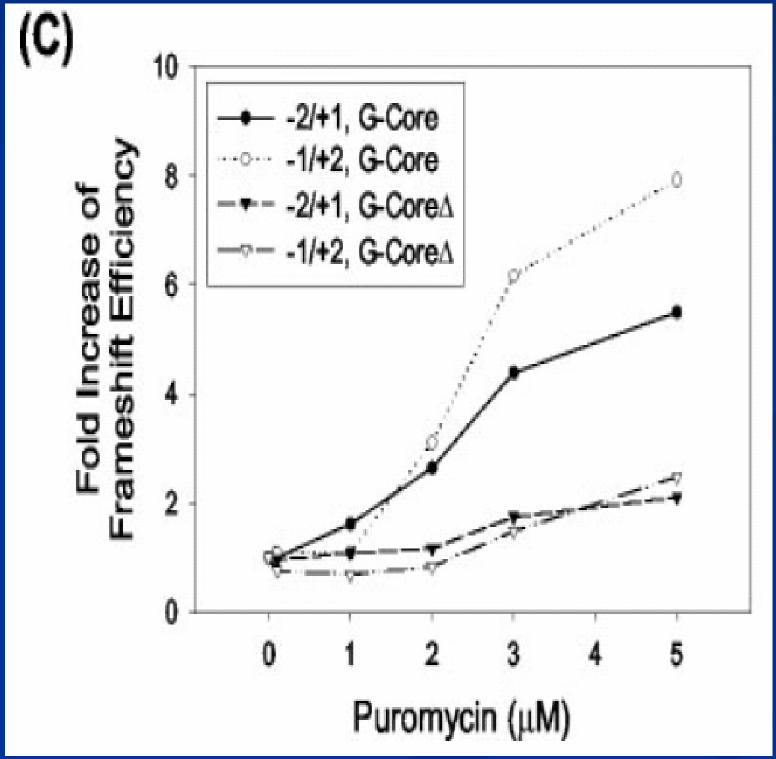
➔ in good agreement

Enhanced frameshift by stem-loop

- Flanking region : NO apparent effect on frameshift (Xu *et al*, 2001)
- Codons 15 to 53 : putative double stem-loop structure
  - Possible effect in the presence of other regulators
- Puromycin : Elongation inhibitor that is known to induce frameshift.

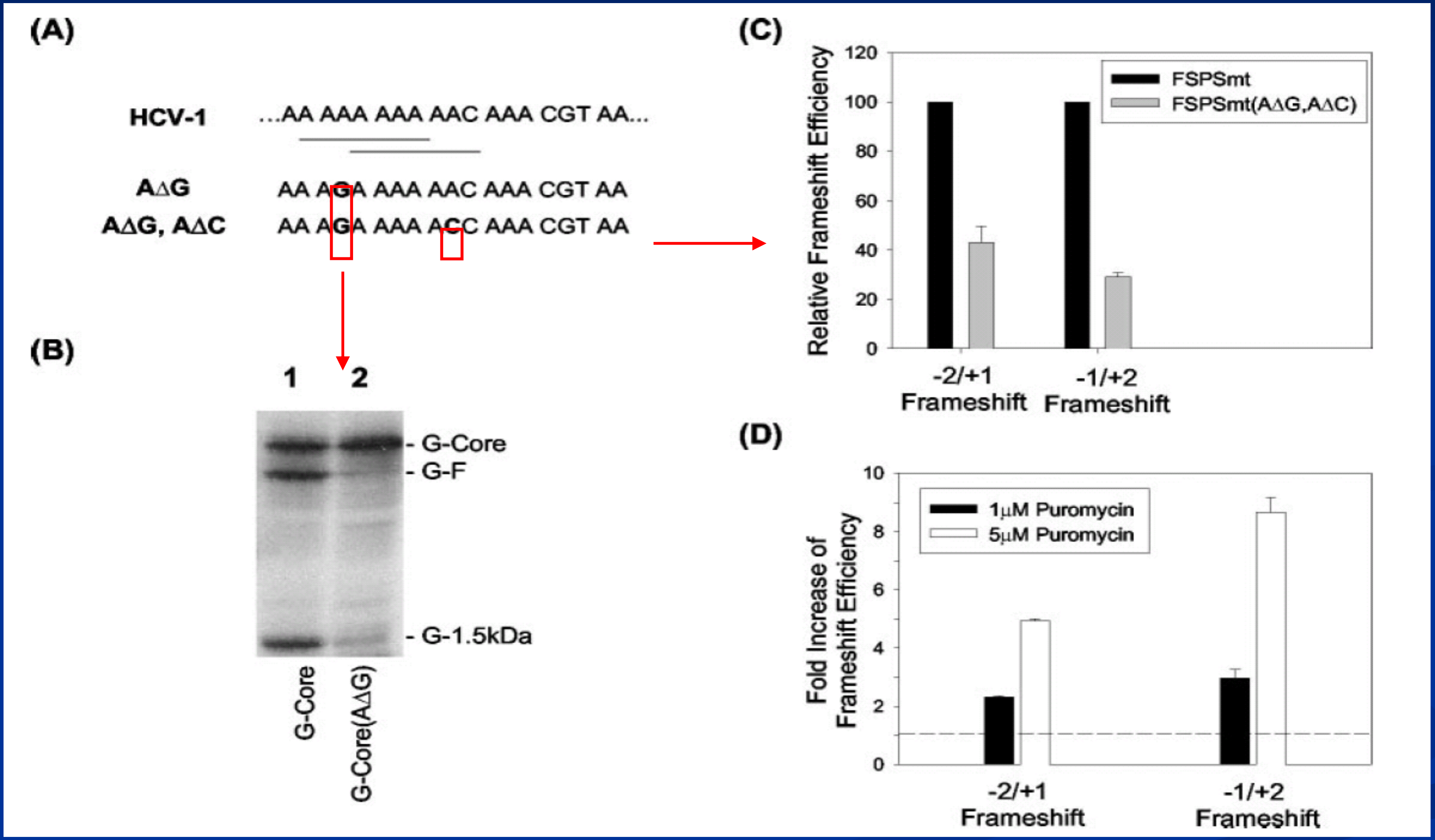
Enhanced frameshift by stem-loop





PS (Putative stem) enhanced frameshift in dose-dependent manner

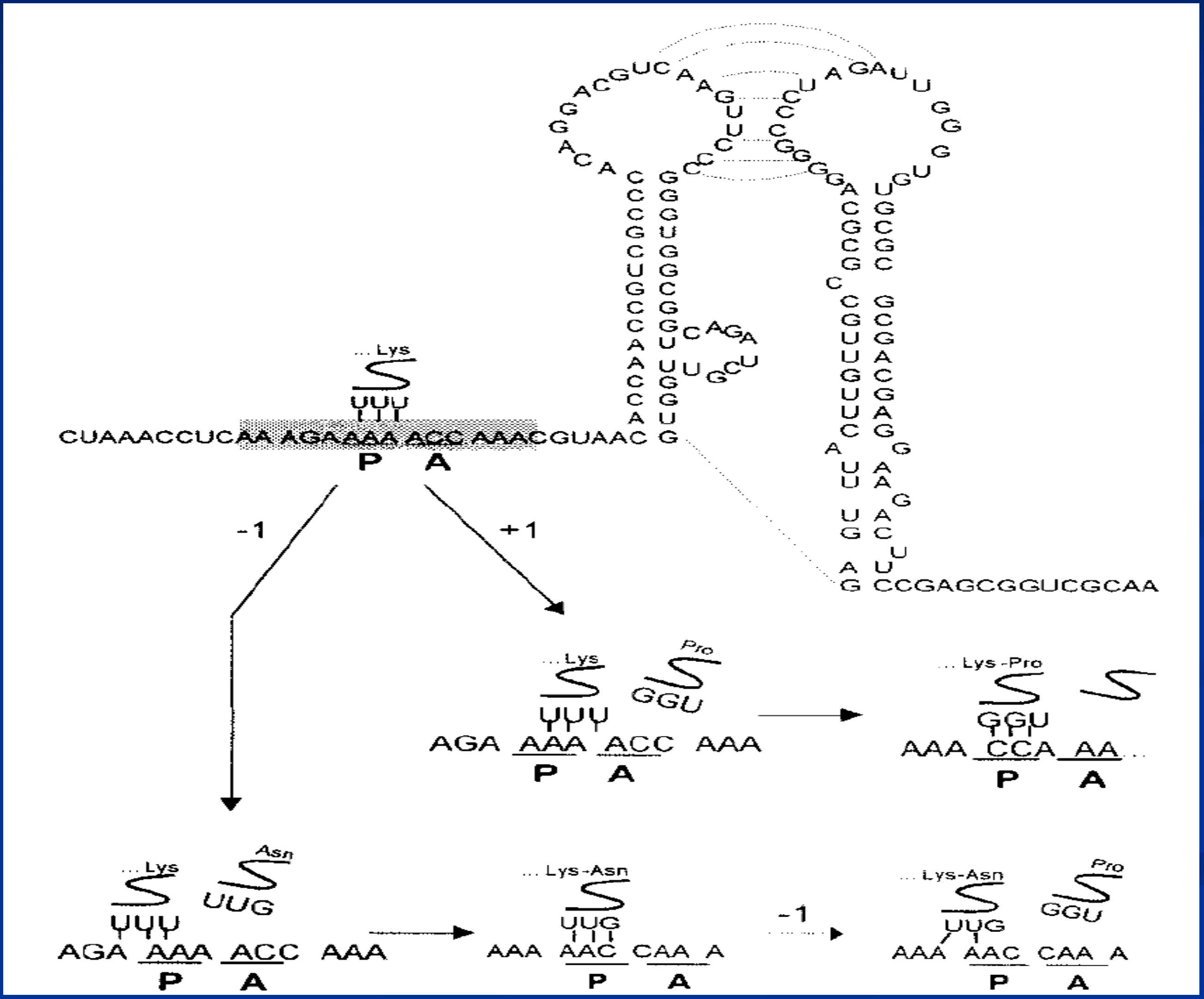
Double Mutation



No consensus → still -1 and +1 frameshift happen

➔ New proposed mechanism : slippage of tRNA at codon 10 by +1 / -1

Mechanism of frameshift in the absence of consensus signal



Additional Claims

- Double mutation of A to G at nt 376 and A to C at nt 382
  - Large fraction of HCV sequences
- Dual frameshifting, although at efficiencies lower than those of the HCV-1 sequence → countered by the next paper
- Possible pseudoknot from the double stem-loop structure

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1474–1486 *Nucleic Acids Research, 2005, Vol. 33, No. 5*
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Translation of the F protein of hepatitis C virus is initiated at a non-AUG codon in a +1 reading frame relative to the polyprotein

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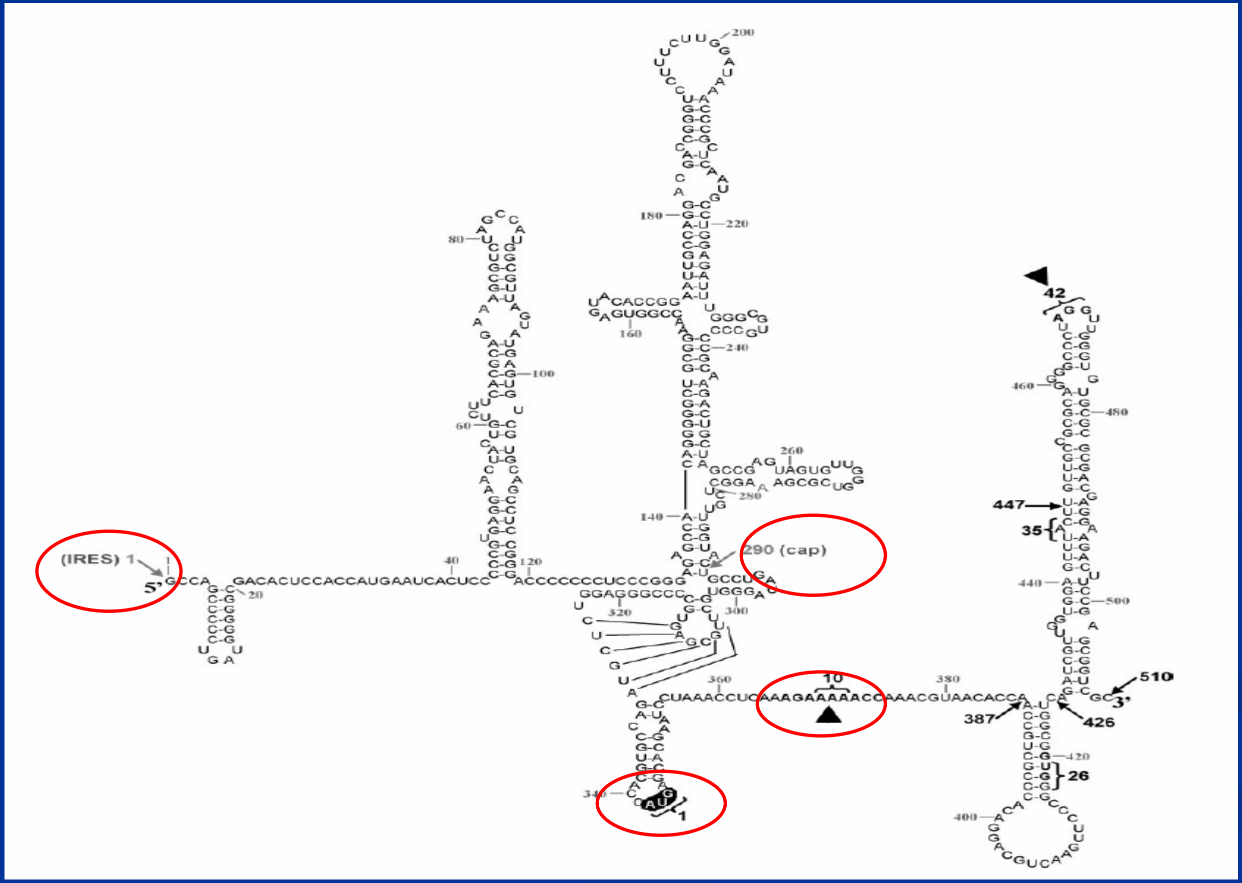
Received January 21, 2005; Revised and Accepted February 20, 2005

Motivation

- A stretch of 10 adenines (10A) encompassing the proposed frameshift site → underrepresented among the viral variants (only 2 of 721 sequences)
- High efficiency *in vitro* but less in cultured mammalian cells
← stringent control of translation
- Antibody against F protein can be detected in patients infected with any HCV genotype, whether 10A stretch is present or not

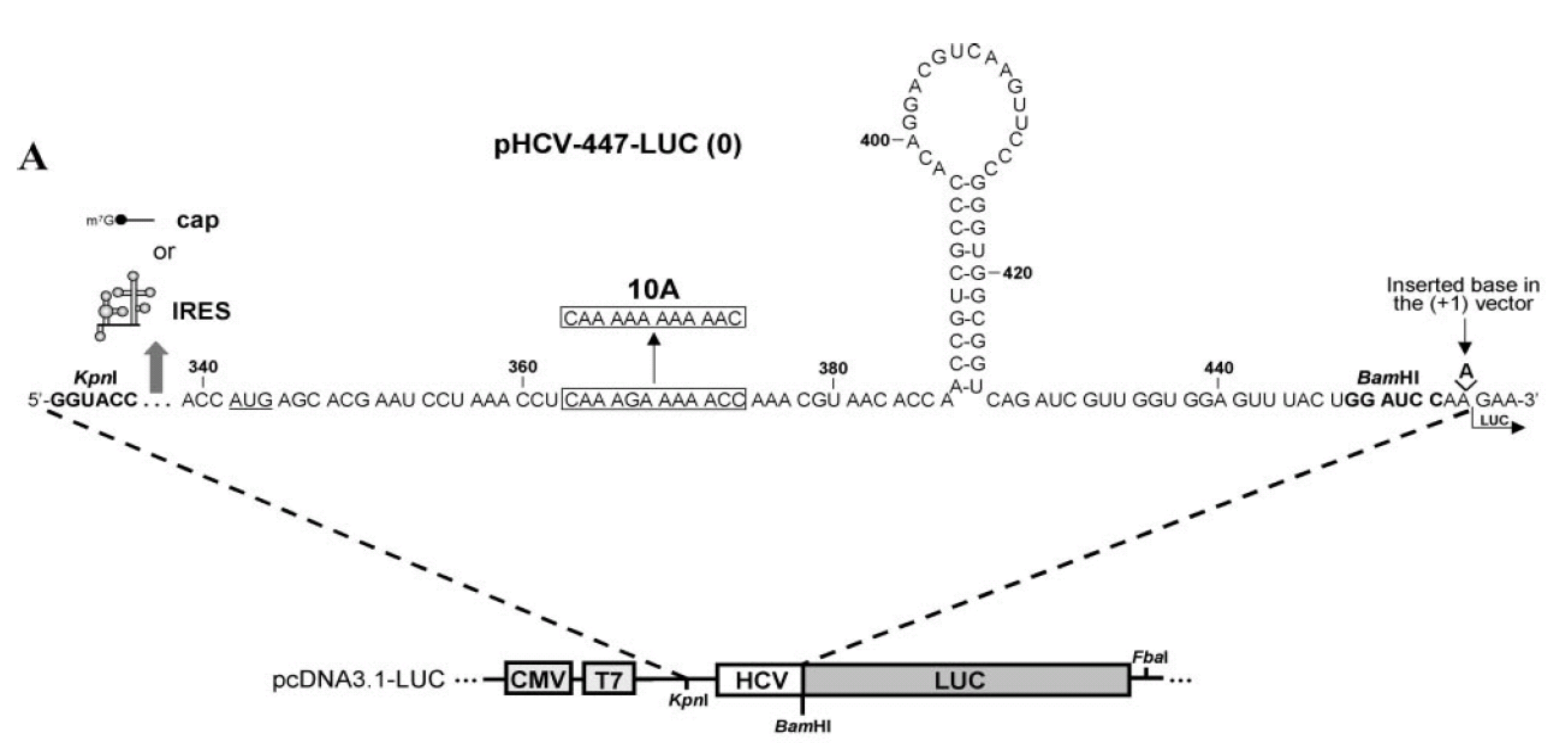
Methods

- Different length of HCV RNA + firefly luciferase gene



Two different types
1) IRES
2) cap

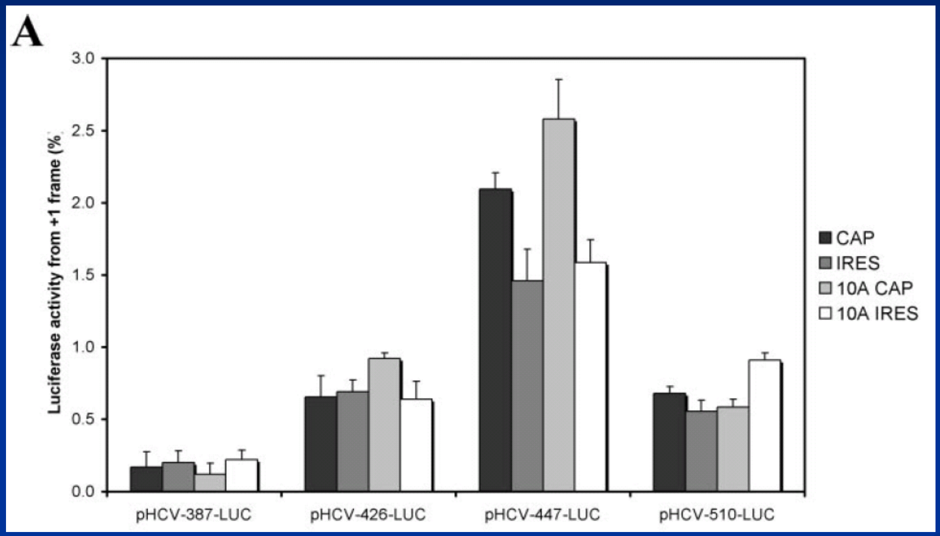
Luciferase Vector



(+1) construct : made by adding A(denine) before the *luc* coding sequence

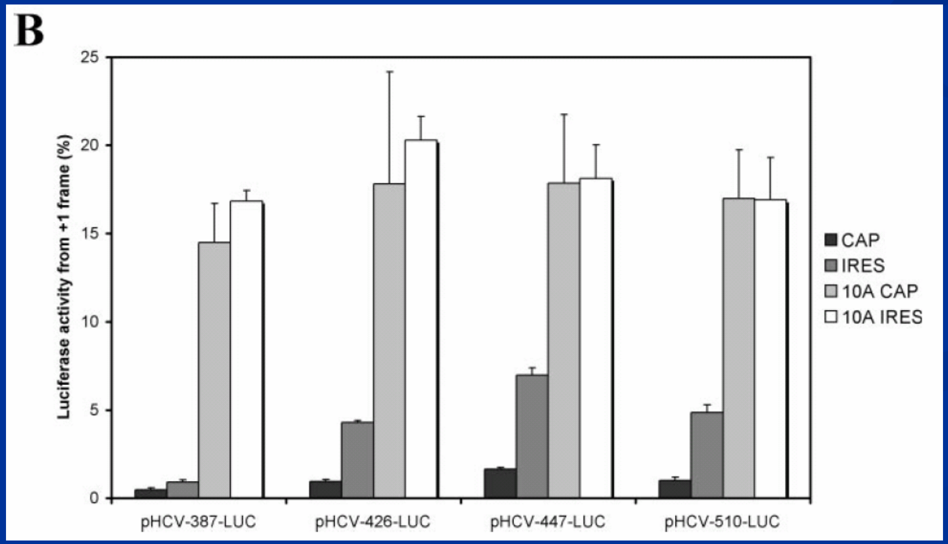
Coding Region for the Maximal F Protein

In vivo



- Background Activity
  - *In vivo*: ~0.4%
  - *In vitro*: ~1%
- Lengthening increased the activity.
- Cap, IRES independent
- Activity value is low but constant (5-fold)
- 10A does not enhance

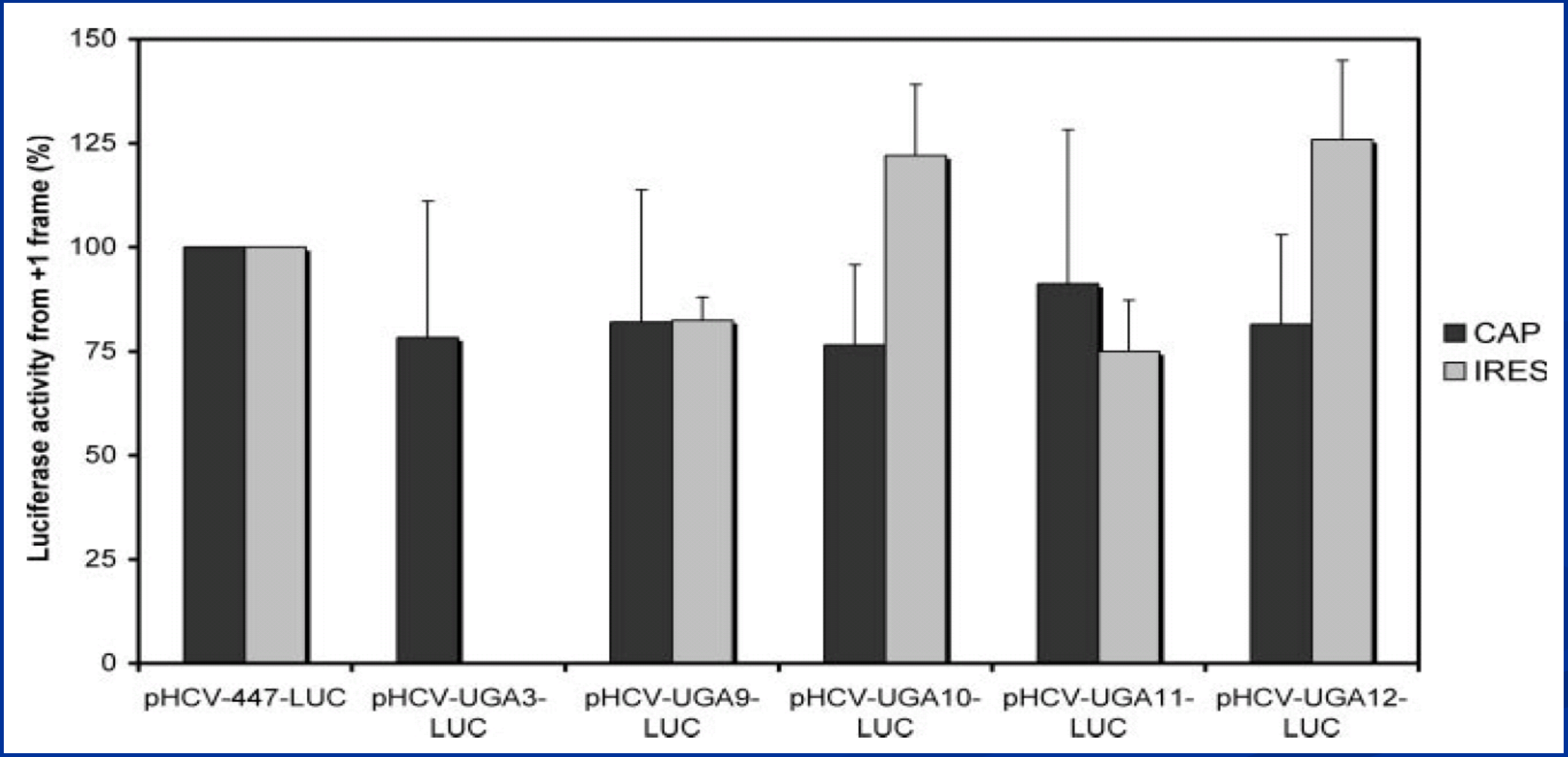
In vitro



- 10A DID promote frameshift, favors a +1 frameshift.
- In vitro translational systems are less accurate than cultured cells

Stop codon at various position

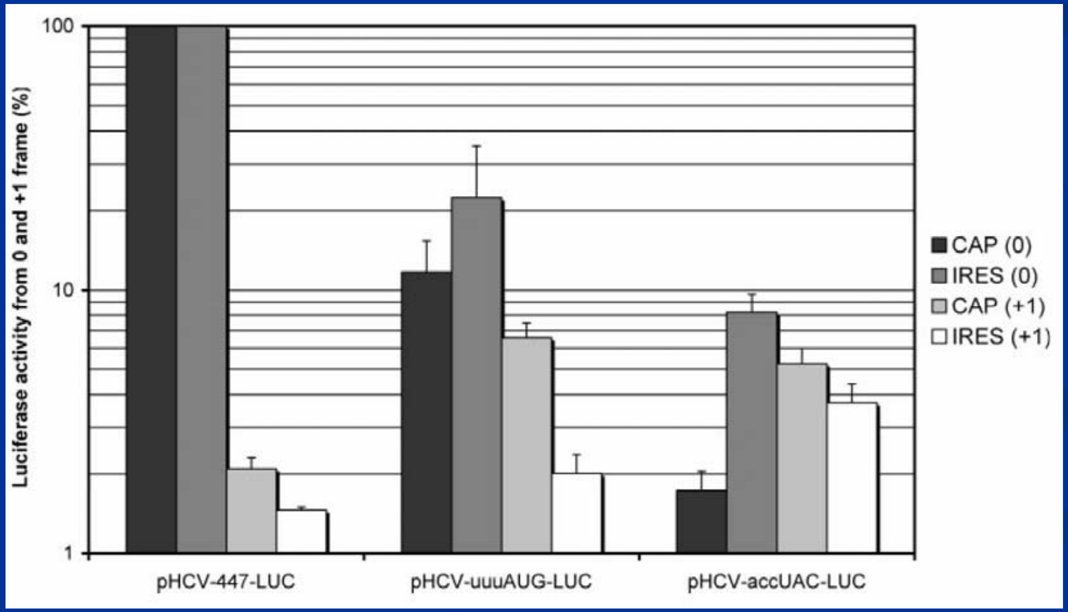
- Introduced stop codon at various position after AUG



- Stop codon did not abolish F protein ➔ F protein is not by frameshift

Efficiency of Initiation of Translation

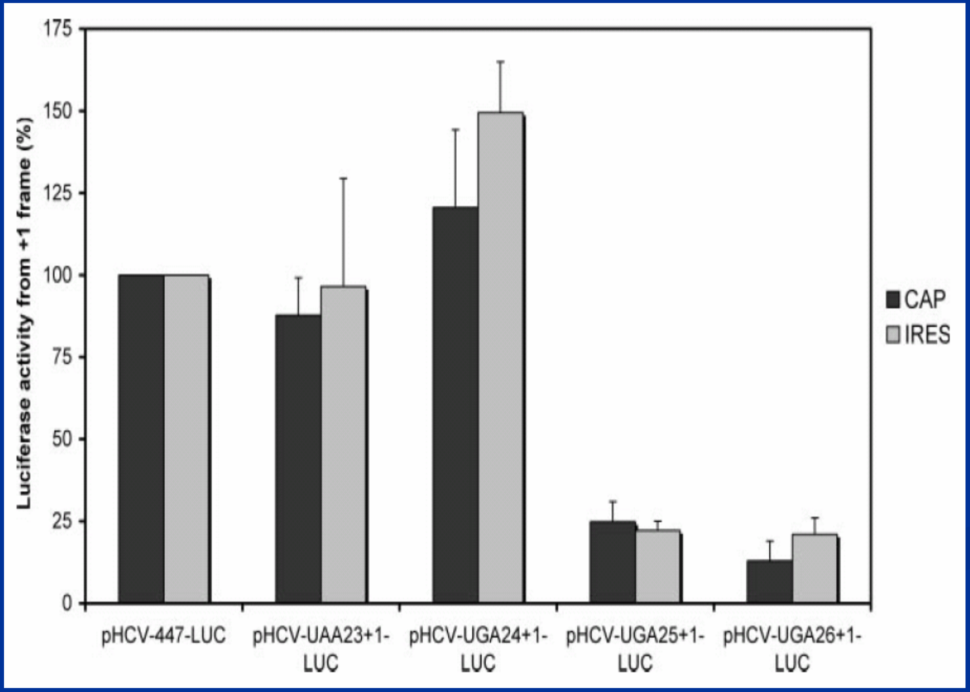
- Previous result → F protein starts at +1 frame initiation site
- Then TSS for the polyprotein and F protein must compete for the amount of available 40S ribosomal subunits.
- Two variants of pHCV-447-LUC
  - pHCV-uuuAUG-LUC : From optimal context -> weak
  - pHCV-accUAC-LUC : Initiator codon was mutated



- polyprotein → decreased
- F protein -> increased

Location of F protein TSS

- Then, where is the TSS for F protein?
- Introduce stop codon at various positions in +1 reading frame

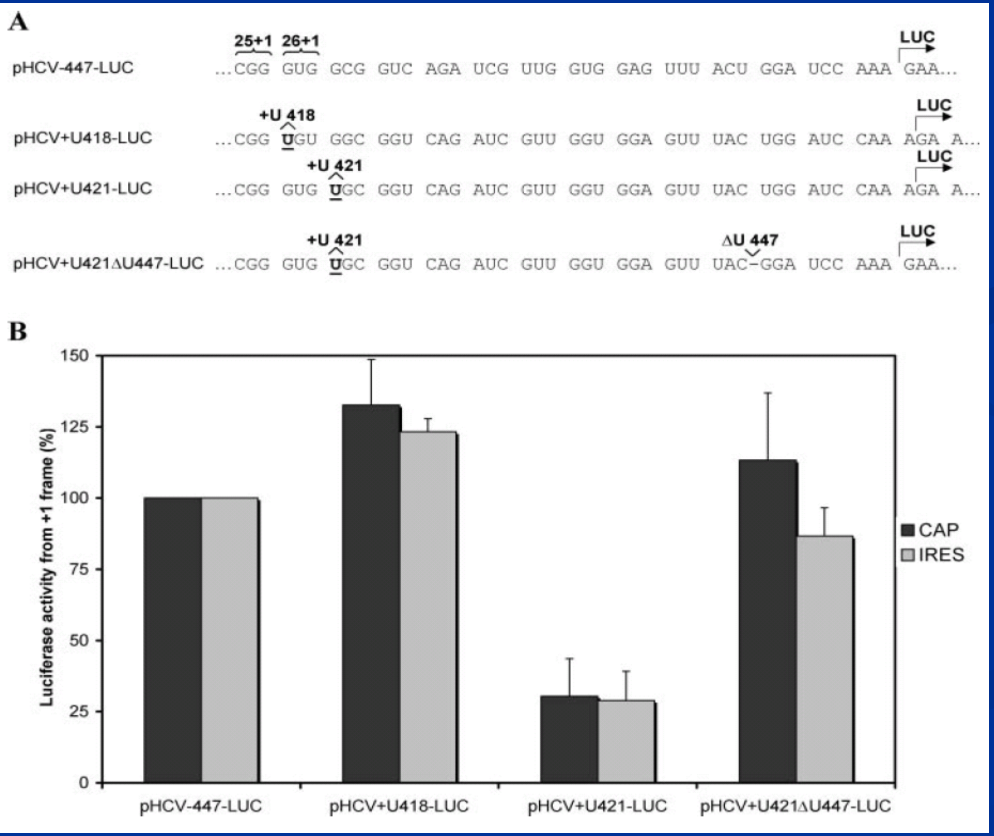


Possible TSS at codon 25 (CGG)

But authors also proposed the possibility of TSS at codon 26 (GUG)

Determination of TSS

- Add additional nucleotide (U) immediately after codon 25(+1) and codon 26(+1) ➔ This will change the reading frame

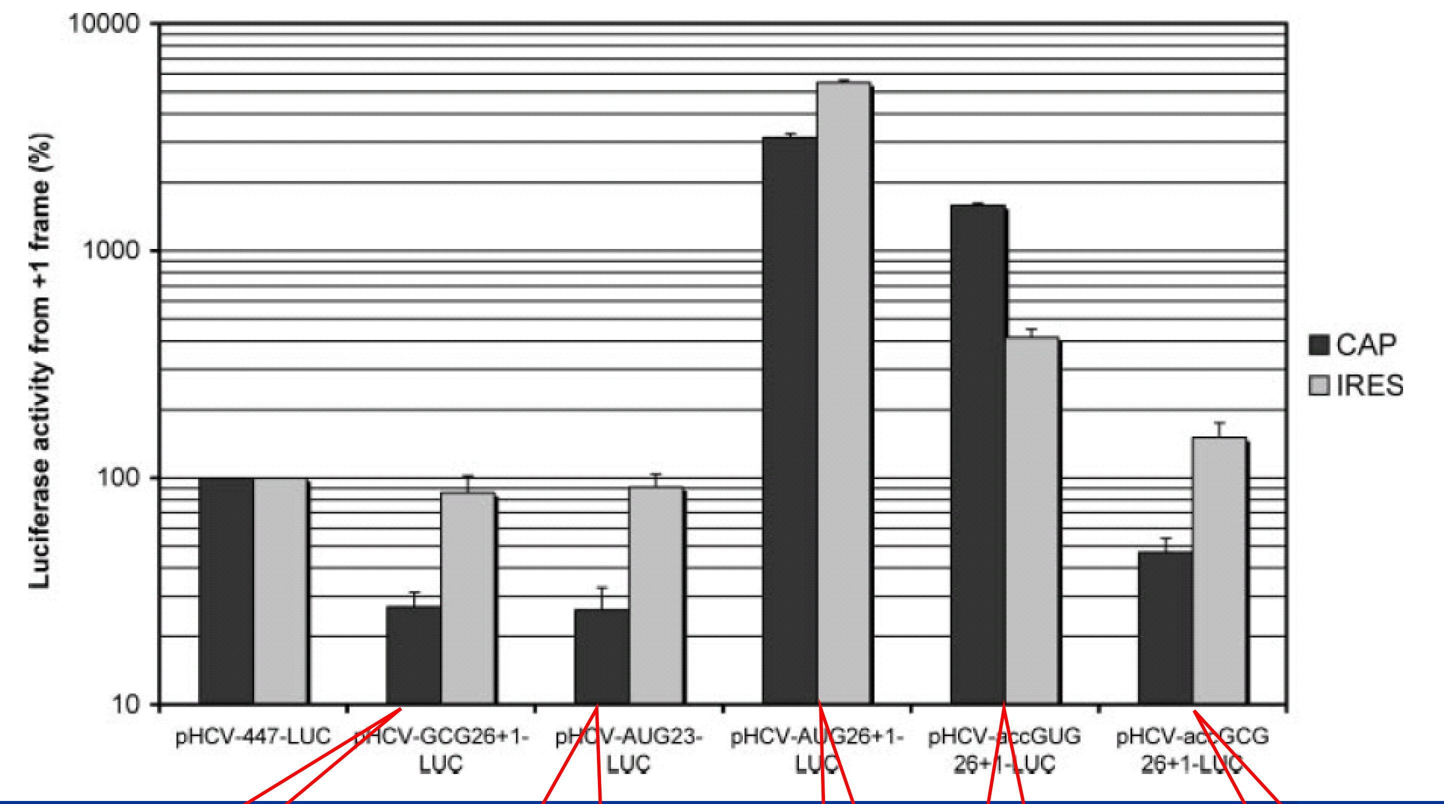


TSS is at codon 26(+1)

Further Information

- Genomic constitution
  - GUG : 18% of the HCV sequences
  - GCG : 81% "
- Binding of S40 to TSS
  - Cap-dependent → leaky scanning
  - IRES-dependent → direct binding

Further Information



GUG -> GCG Decreased activity for CAP

AUG at 23 Decreased activity for CAP – Scanning

AUG at 26 Highly active

Optimal Kozak context For GUG

Optimal Kozak context For GCG

Conclusion of Baril and Brakier-Gingras

- F-protein by frameshift +1 over a stretch of 10A *in vitro*, was using a **underrepresented** shifty sequence (Consensus: *AAAGAAAAAC*)
- *In vivo* and *in vivo* are different.
- Initiation of F protein synthesis starts at codon 26 which is non-AUG

FINAL CONCLUSION

- Controversy exists whether F protein is produced by frameshift or from non-AUG initiation codon
- But I think that frameshift may explain some remaining portion of the F proteins when they are supposed to be impaired by experimental design
- Much work needs to be done

Thank You